WORKSHEET

Chapter 9: Areas between curves

7	13	15	15	4	3	11	14	10	14	8	15	5		

12	10	14	9	13	12	6	2	14	1					

Saint Francis de Sales (1567–1622)

Questions

For $y = -x^2 + 7x - 6$ and $y = x + 2$:

1 Sketch the graphs on the same set of axes.

2 At what values of x do these two graphs intersect?

3 Calculate the enclosed area between the curves.

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For $y = x^2 - 4x + 3$ and $y = 2x - 2$:

4 Sketch the graphs on the same set of axes.

5 At what values of x do these two graphs intersect?

6 Calculate the enclosed area between the curves.

For $y = 4x - x^2$ and $y = x^2 - 6x + 12$:

7 Sketch the graphs on the same set of axes.

8 At what values of x do these two graphs intersect?

9 Calculate the enclosed area between the curves.

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For $y = x(x - 3)(x - 5)$ and $y = 3x$:

10 Sketch the graphs on the same set of axes.

11 At what values of x do these two graphs intersect?

12 Calculate the enclosed area between the two curves.

For $y = (x - 2)(x + 1)(x + 5)$ and $y = -8x - 4$:

13 Sketch the graphs on the same set of axes.

14 At what values of x do these two graphs intersect?

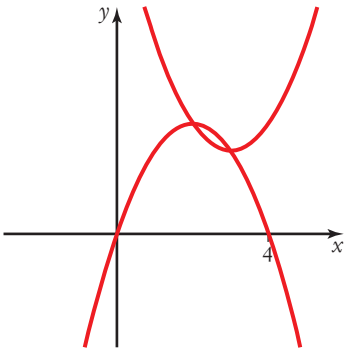
15 Calculate the enclosed area between the curves.

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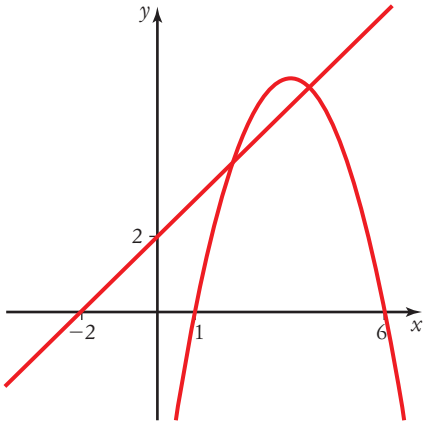
Solutions

A 49.33 units^2

B



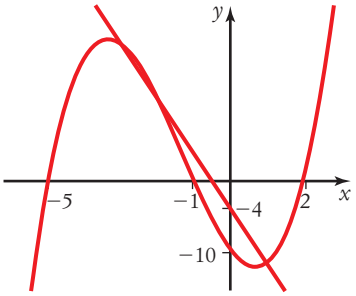
D



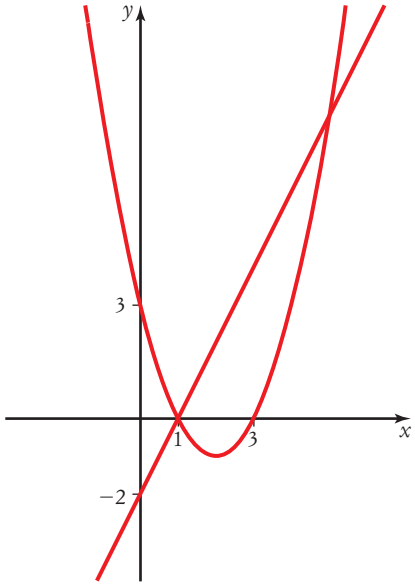
E $x = -3$, $x = -2$ and $x = 1$

H $x = 0$, $x = 2$ and $x = 6$

L



M

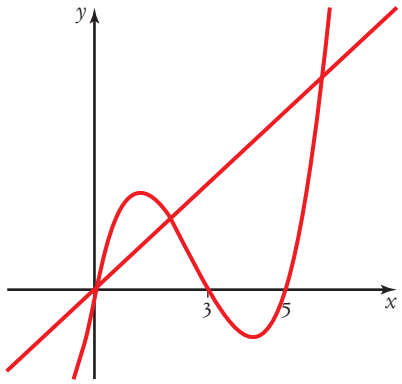


N 10.67 units^2

O 11.83 units^2

P 0.33 units^2

R



T $x = 2$, and $x = 4$

U $x = 1$ and $x = 5$

W 1.3 units^2

Y $x = 2$ and $x = 3$

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